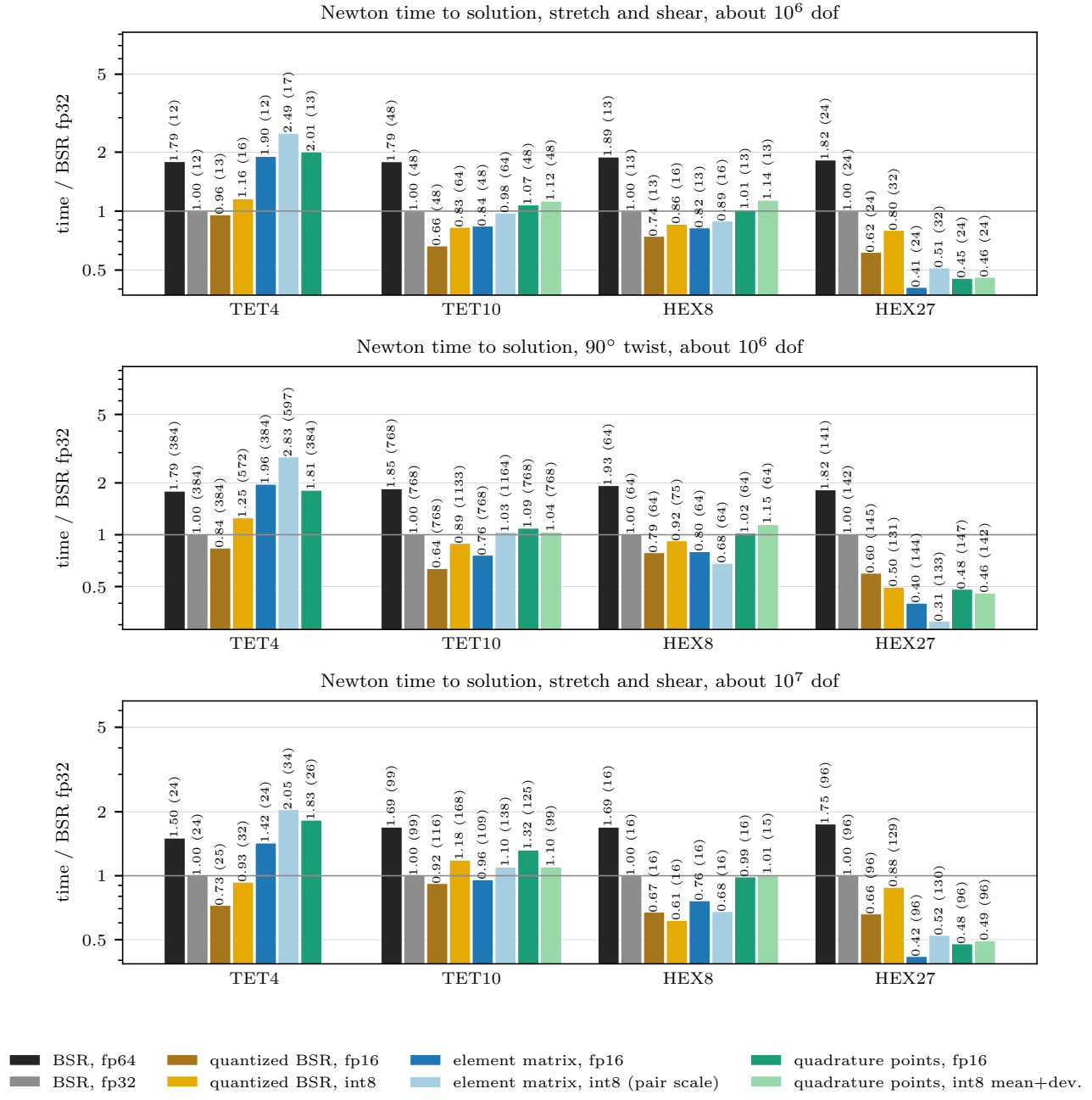

Newton time to solution, stretch and shear, about 10^6 dof
time / BSR fp32
5
2
1
0.5
1.79 (12)
1.00 (12)
0.96 (13)
1.16 (16)
1.90 (12)
2.49 (17)
2.01 (13)
1.79 (48)
1.00 (48)
0.66 (48)
0.83 (64)
0.84 (48)
0.98 (64)
1.07 (48)
1.12 (48)
1.89 (13)
1.00 (13)
0.74 (13)
0.86 (16)
0.82 (13)
0.89 (16)
1.01 (13)
1.14 (13)
1.82 (24)
1.00 (24)
0.62 (24)
0.80 (32)
0.41 (24)
0.51 (32)
0.45 (24)
0.46 (24)
TET4
TET10
HEX8
HEX27
Newton time to solution, 90° twist, about 10^6 dof
time / BSR fp32
5
2
1
0.5
1.79 (384)
1.00 (384)
0.84 (384)
1.25 (572)
1.96 (384)
2.83 (597)
1.81 (384)
1.85 (768)
1.00 (768)
0.64 (768)
0.89 (1133)
0.76 (768)
1.03 (1164)
1.09 (768)
1.04 (768)
1.93 (64)
1.00 (64)
0.79 (64)
0.92 (75)
0.80 (64)
0.68 (64)
1.02 (64)
1.15 (64)
1.82 (141)
1.00 (142)
0.60 (145)
0.50 (131)
0.40 (144)
0.31 (133)
0.48 (147)
0.46 (142)
TET4
TET10
HEX8
HEX27
Newton time to solution, stretch and shear, about 10^7 dof
time / BSR fp32
5
2
1
0.5
1.50 (24)
1.00 (24)
0.73 (25)
0.93 (32)
1.42 (24)
2.05 (34)
1.83 (26)
1.69 (99)
1.00 (99)
0.92 (116)
1.18 (168)
0.96 (109)
1.10 (138)
1.32 (125)
1.10 (99)
1.69 (16)
1.00 (16)
0.67 (16)
0.61 (16)
0.76 (16)
0.68 (16)
0.99 (16)
1.01 (15)
1.75 (96)
1.00 (96)
0.66 (96)
0.88 (129)
0.42 (96)
0.52 (130)
0.48 (96)
0.49 (96)
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
quantized BSR, fp16
quantized BSR, int8
element matrix, fp16
element matrix, int8 (pair scale)
quadrature points, fp16
quadrature points, int8 mean+dev.